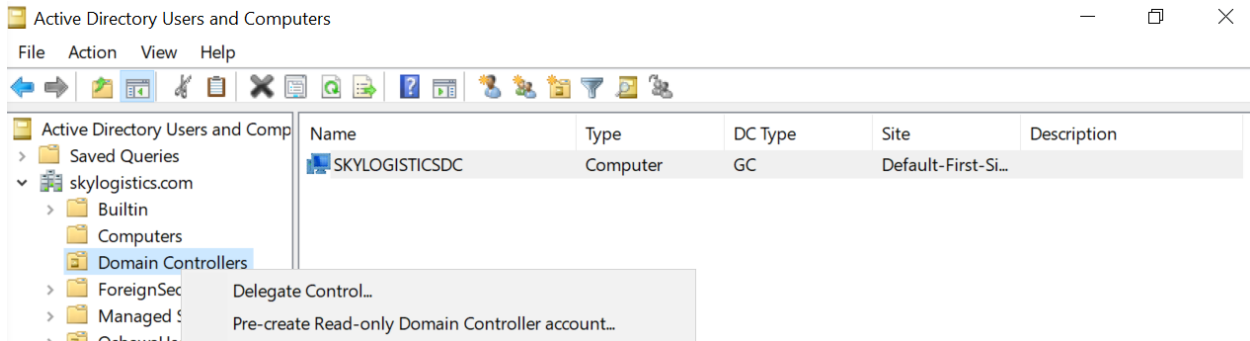


Week 6 Lab: Advanced Active Directory Configuration

Using the primary DC for your company, you will configure your Active Directory environment using some advanced features. Include screenshots of each outlined step. Use cropping to include only relevant information in your screenshots. Uncropped screenshots will result in a deduction of 10%.

Part 1: Sites, Subnets, and Replication

1. Pre-create an RODC account that can be used for the deployment of a DC at a branch office.



Welcome to the Active Directory Domain Services Installation Wizard

This wizard helps you create an account for a read-only domain controller (RODC). You will be able to attach the server that you want to be the RODC to this account by running this wizard on that server.

☒ Use advanced mode installation

Computer name:

SKYRODC

Full DNS computer name:

SKYRODC.skylogistics.com

2. Configure the Password Replication policy on your RODC with the settings of your choosing.
Take a screenshot of the policy and explain your choices.

Specify the Password Replication Policy

In the Password Replication Policy, you can specify whose passwords are allowed to replicate to the read-only domain controller (RODC) or whose accounts are denied from replicating to the RODC.



Add or remove any accounts whose passwords you want to allow to replicate to this RODC or whose passwords you want to deny from replication to this RODC.

Groups, users and computers:

Name	Active Directory Domain Servic...	Setting	
Denied RODC Passwor...	skylogistics.com/Users	Deny	Add... Remove
Allowed RODC Passwor...	skylogistics.com/Users	Allow	
Daniel	skylogistics.com/OshawaUsers/...	Allow	
Tim	skylogistics.com/OshawaUsers/...	Allow	
Melvin	skylogistics.com/OshawaUsers/...	Allow	

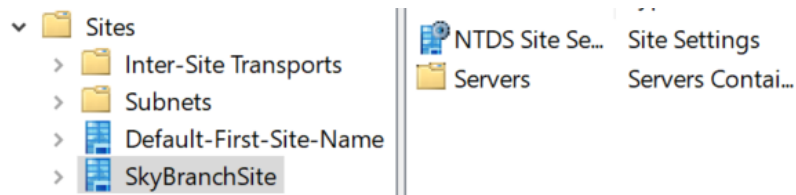
For the specification of the password replication policy, I kept the default Denied RODC Password Replication Group because it had already contained all the privileged groups as it is not advised to have their credentials cached on the RODC. Malicious actors that want to compromise the RODC will expose very privileged accounts.

I allowed three users in the Oshawa branch office because those are the members that work at the site. This will be the allowed RODC Password Replication Group which are not global groups like "Domain Users" with the aim to not include unnecessary credential exposure.

Group or user:

SKYLOGISTICS\DanielAdmin	Set...
--------------------------	--------

3. Create a new site for a branch office at your company.
 - a. If your company doesn't have a branch office, use this site as a backup site



- Assign the site created in step 3 to a subnet. Take a screenshot of the subnet properties to show the assigned site.

New Object - Subnet



 Create in: skylogistics.com/Configuration/Sites/Subnets

Enter the address prefix using network prefix notation (address/prefix length), where the prefix length indicates the number of fixed bits. You can enter either an IPv4 or an IPv6 subnet prefix.
[Learn more about entering address prefixes.](#)

IPv4 example: 157.54.208.0/20

IPv6 example: 3FFE:FFFF:0:C000::/64

Prefix:

Prefix name in Active Directory Domain Services:

Select a site object for this prefix.

Site Name

	Default-First-Site-Name
	SkyBranchSite

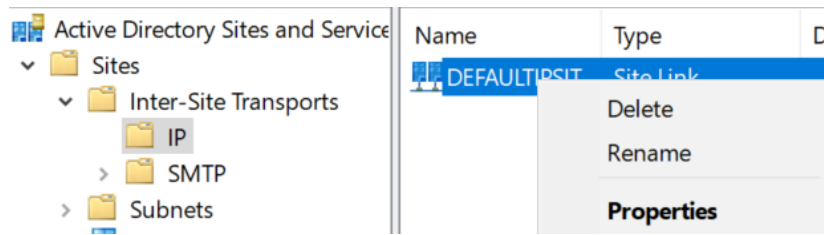
- Enable Universal Group Membership Caching for the created site.

Universal Group Membership Caching

☒ Enable Universal Group Membership Caching

Refresh cache from:

6. Change the DEAFULTIPSITELINK replication schedule to replicate every **240 minutes**.



Replicate every minutes

Part 2: ADCS

1. Create a new VM running Windows Server 2019. Add it to your organization's domain.
 - a. Remember to give it an appropriate name and static IP

☒ Use the following IP address:

IP address:	172 . 10 . 0 . 88
Subnet mask:	255 . 255 . 0 . 0
Default gateway:	172 . 10 . 0 . 2

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:	172 . 10 . 0 . 85
-----------------------	--

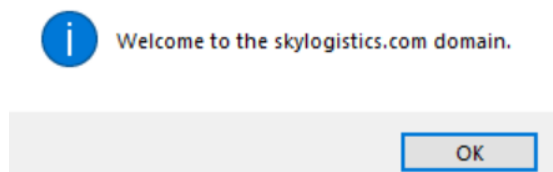
Computer name:

Full computer name:
SkyLogisticsCA

Member of

☒ Domain:

Computer Name/Domain Changes



2. Promote the server to a Certificate Authority server by installing the AD CS role
 - a. Include a screenshot of the confirmation screen before completing the role installation process

View installation progress



Feature installation

Configuration required. Installation succeeded on SkyLogisticsCA.skylogistics.com.

Active Directory Certificate Services

Additional steps are required to configure Active Directory Certificate Services on the destination server

[Configure Active Directory Certificate Services on the destination server](#)

Certification Authority

Remote Server Administration Tools

Role Administration Tools

Active Directory Certificate Services Tools

Certification Authority Management Tools

- b. Include a screenshot of the AD CS role configuration confirmation as well

Confirmation

DESTINATION SERVER

SkyLogisticsCA.skylogistics.com

Credentials

Role Services

Setup Type

CA Type

Private Key

Cryptography

CA Name

Validity Period

Certificate Database

Confirmation

Progress

Results

To configure the following roles, role services, or features, click Configure.



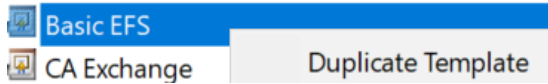
Active Directory Certificate Services

Certification Authority

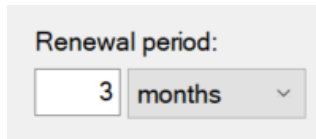
CA Type:	Enterprise Root
Cryptographic provider:	RSA#Microsoft Software Key Storage Provider
Hash Algorithm:	SHA256
Key Length:	2048
Allow Administrator Interaction:	Disabled
Certificate Validity Period:	3/3/2031 6:41:00 AM
Distinguished Name:	CN=skylogistics-SKYLOGISTICSCA-CA-3,DC=skylogistics,DC=com
Certificate Database Location:	C:\Windows\system32\CertLog
Certificate Database Log Location:	C:\Windows\system32\CertLog

3. Configure your CA (Enterprise)

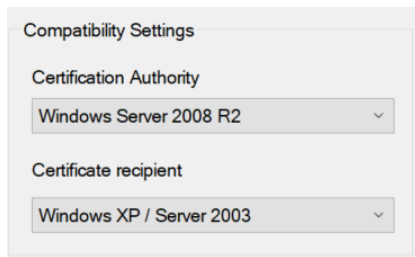
- a. Duplicate the Basic EFS Template and make the following adjustments:



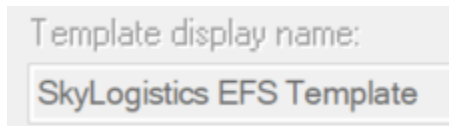
- i. Renewal Period: 3 months



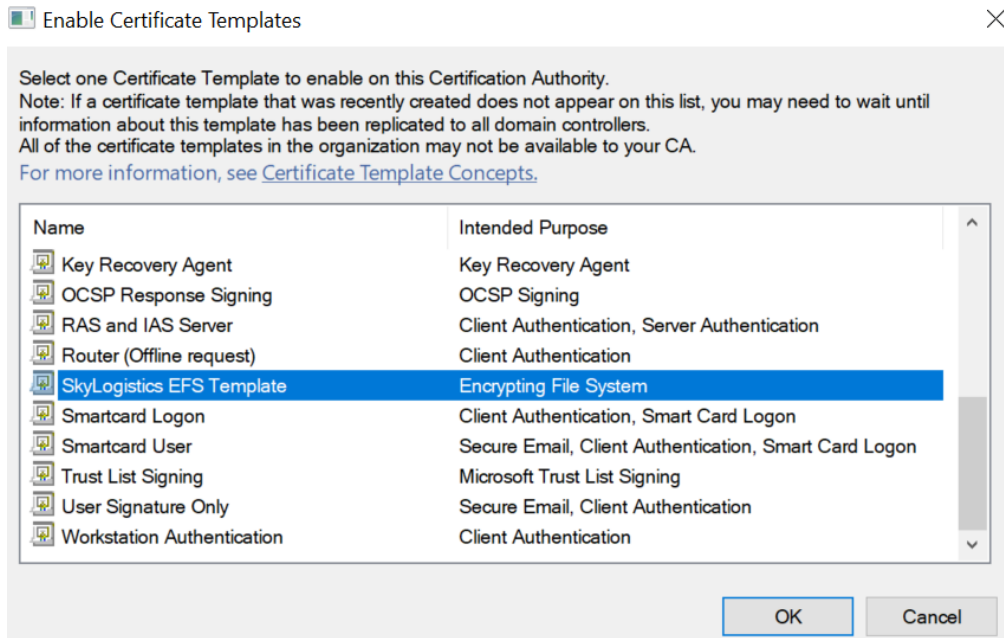
- ii. Compatibility CA: Windows Server 2008 R2



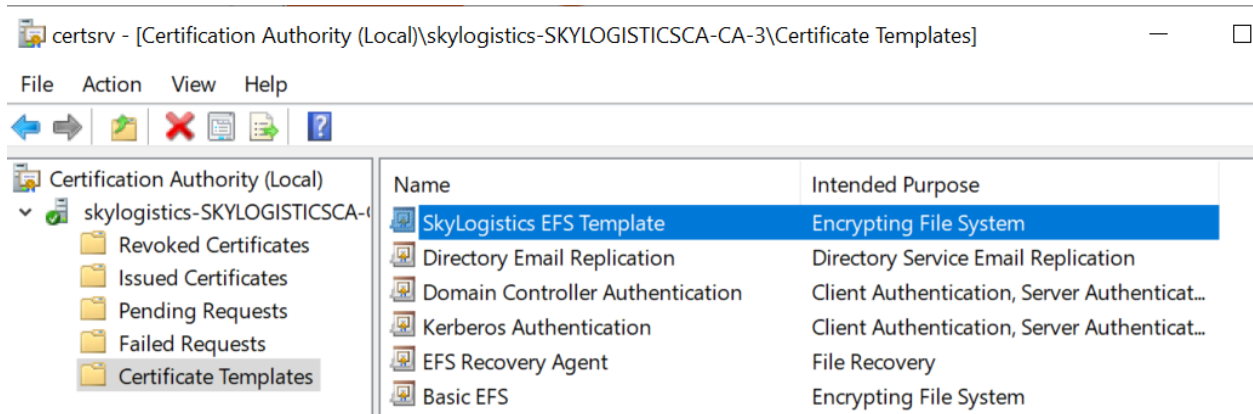
- iii. Name: <Organization Name> EFS Template



- b. Add the new certificate template to your Certificate Template store listed in the Certification Authority tool



- i. Take a screenshot of the new certificate listed under Certificate Templates in the Certification Authority tool



4. Create a domain-joined user computer and request a certificate from the newly created certificate template
 - a. Take a screenshot of the user's issued certificate

```
C:\Users\melvin>certreq -enroll -q -user "SkyLogisticsEFSTemplate"
Active Directory Enrollment Policy
{4857042E-77C2-4BD5-B6C7-84280D980F4E}
ldap:
The operation completed successfully.
RequestId = 2
SkyLogisticsCA.skylogistics.com\skylogistics-SKYLOGISTICSCA-CA-3
Serial Number: 1600000002efdb38482e3953b4000000000002
638b23e8f38aa054db337ecbb80965b6f1fc8dc4
Key Container Name: 6ab84551b8f8baa35f0336b66cf7eddf_db72567b-ac94-4a81-99a4-a5a456c68192
Key Container Name: te-SkyLogisticsEFSTemplate-42447601-de41-42e9-97ad-b8cc245985f5

The requested certificate has been issued.
```